

MUGUNTHAN KESAVAN

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PROFESSIONAL SUMMARY

AI/ML Engineer with expertise in deep learning, NLP, computer vision, and LLM deployment. Experienced building end-to-end ML pipelines, RAG systems, and full-stack applications using Python, PyTorch, TensorFlow, and cloud infrastructure. Published researcher with 2 peer-reviewed papers and strong technical writing skills. Proficient in MLOps, statistical analysis, data visualization, project management, and agile development.

TECHNICAL SKILLS

Languages: Python, SQL, R, Scala, Java, C++, JavaScript | **Frameworks:** PyTorch, TensorFlow, Hugging Face, scikit-learn, Spark MLlib, FastAPI, React, OpenCV

AI/ML: Deep Learning, NLP, Computer Vision, LLMs (GPT, LLaMA, BERT, Phi-3, Qwen, Mistral), RAG, Transformers, CNNs, RNNs, XGBoost, Transfer Learning

Tools & Infra: Airflow, Spark, Hadoop, Docker, Git, CI/CD, REST APIs, PostgreSQL, MongoDB, SQLite, AWS, GCP, ETL Pipelines, Feature Engineering | **Analysis & Viz:** Matplotlib, Seaborn, Pandas, NumPy, SciPy, SPSS, Tableau, Power BI, MATLAB, Data Visualization, Statistical Software, Microsoft Office Suite | **Other:** Technical Writing, Project Management, Research Methodologies, Teamwork, Organizational Skills, Agile Development

WORK EXPERIENCE

Technical Research Assistant — AI Systems & Automation

Texas Manufacturing Assistance Center (TMAC) x University of Houston, Houston, TX Mar 2025 – Present

- Engineered a multi-LLM RAG pipeline (Phi-3 Mini, Qwen 2.5, Mistral) to automate order validation across ~800 emails/month, achieving 97% classification accuracy against NetSuite ERP records.
- Led project management and organizational planning of end-to-end pipeline orchestration with Apache Airflow, MongoDB, and PostgreSQL for automated ingestion, extraction, and validation.
- Implemented NLP-based email parsing to extract structured order fields, reducing manual data entry by 85% and cutting per-order processing time significantly.

Research Assistant — Digital Twin & Simulation

Hewlett Packard Enterprise (HPE) x University of Houston, Houston, TX Jan 2026 – May 2026

- Built a full-stack digital twin of an 8-station reverse supply chain using Python FastAPI, SimPy, React 19, and Three.js for real-time 3D visualization.
- Automated data entry via document parsing, reducing per-unit processing time from 20 min to 2 min (90% reduction); implemented QR-based throughput tracking.
- Developed data visualization dashboard with what-if simulation and Ollama (Phi-3 Mini) LLM integration for natural language Q&A; collaborated cross-functionally with HPE engineering leadership.

Research Intern — Machine Learning

Flame University, Pune, India Jan 2025 – Jul 2025

- Conducted statistical analysis using SciPy and scikit-learn on 30K+ protein sequences; applied research methodologies including 10-fold cross-validation to train ProtBERT/ESM2 + SVM models achieving 98.3% accuracy and 98% F1-score.
- Authored technical writing for peer-reviewed Springer publication on ML framework for protein prediction; automated data preprocessing with Python/Pandas, reducing manual effort by 50%.

EDUCATION

M.S., Engineering Data Science and Artificial Intelligence | GPA: 3.8/4.0 Aug 2025 – May 2027

University of Houston | Coursework: ML, Data Science, Statistics, Neural Networks, Biomedical Data Mining, Predictive Analytics

B.Tech., Artificial Intelligence Engineering | GPA: 3.7/4.0 Aug 2021 – May 2025

Amrita Vishwa Vidyapeetham | Coursework: ML, Deep Learning, Big Data Analytics, NLP, Distributed Systems

SELECTED PROJECTS

Neural Style Transfer for Video | Built segmentation-based NST pipeline for real-time video using DeepLabV3 + VGG-19 with independent foreground/background stylization. Published on GitHub.

Sales Analytics Pipeline | Designed distributed pipeline for 15M+ records using Hadoop, MapReduce, and Spark MLlib with data visualization dashboards; achieved 40% computation time reduction vs. serial baseline.

PUBLICATIONS

Rupa, N.B., Hima, Y., **Mugunthan, K.**, & Premjith, B. (2024). *Humor Detection in Telugu Social Media Text Using Cost-Sensitive Learning*. ICRTC, pp. 49-60.

Jain, R., **Kesavan, M.**, & Jayaraman, V.K. (2025). *Comparative Study of Zero-Shot and Supervised Learning for Tamil Tweet Sentiment Analysis*. ICDSA 2025, pp. 85-99.

LEADERSHIP & ACTIVITIES

Led Python/ML/Model Optimization workshops for 70+ students | Mentored 10+ juniors on end-to-end AI/DS projects | Top 20% on Kaggle ML competitions